<u>Hardware Design Engineer Profile</u>

DINESH R
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CAREER OBJECTIVE:

Seeking an opportunity to apply and expand my technical and interpersonal skills in a dynamic environment that fosters professional growth and contributes to the success of the organization.

PROFESSIONAL EXPERIENCE:

Total Experience: 3 Years

Company Name	Designation	From	To
HCL Technologies	Hardware Design Engineer	May 2022	Till Date

PROFESSIONAL SUMMARY:

- Experience in Automated Test Equipment (ATE) development, Hardware design, testing and validation for Unit under Test (UUT) across aerospace, automotive and Industrial domains.
- Skilled in end-to-end ATE development, including requirement analysis, components and COTS selection, Bill of Materials (BOM) creation, Hardware Design & Integration, Electrical Harness Design, Functional Testing & Verification (ATP/DVT).
- Experience in AC Power Distribution Unit (PDU) Design for ATE, Bed of Nail Fixtures Design and Testing.
- Basic knowledge in PLC Hardware.

TECHNICAL SKILS:

Design Tools	Altium Designer schematics, MS Visio
Simulation Tools	NI MAX, LTspice & TINA TI
Software Tools (Basic Knowledge)	C programming & NI LabVIEW

ACADEMIC QUALIFICATION:

- Bachelor of Technology (B. Tech) in Electronics and Communication Engineering, securing 87.6% from Puducherry Technological University (2022).

PROJECTS

i. **BDCU & EMCU ATE**

Client	Parker Meggitt (Aerospace)

Project Description	Design and Development of Automated Test Equipment to perform automated tests to validate the Brake Data concentrator Unit (BDCU) and Electric motor control unit (EMCU) UUT.
Role	Design & Testing
Hardware Used	DSO, DMM, Industrial Power Supply, FG, CAN, NI DAQ, NI Reconfigurable I/O FPGA (USB 7856R)
Responsibilities	○ AC PDU & Harness Design, Interconnection Modules Design, LVDT PCB (Design & Schematic) ○ Components & Equipment selection, BOM Creation, Mechanical & Assembly support, Testing and Validation

ii. Landing Gear control Interface Unit (LGCIU) Tester

Client	CRANE Aerospace (Aerospace)
Project Description	Design and Development of automated tester to validate the LGCIU
Role	Design & Testing
Hardware Used	NI PXI chassis, Self-Test ITA, Analog Current I/O, Discrete I/O, Analog voltage I/O, Temperature modules, Fault Insertion Cards & Multiplexer Cards,
Responsibilities	○ ATE Design & Schematics ○ Component selection, BOM Creation, Mechanical & Supply chain support, Integration Testing and Validation

iii. Vehicle control unit (VCU) Tester

Client	SAIETTA Electric Drive (Automotive)
Project Description	Design and Development of automated test rig to validate the VCU
Role	Design & Testing
Hardware Used	DAQ, High side and low side switch Load PCB & PWM Measurement.
Responsibilities	○ Component selection, BOM Creation, Fixture and Pogo pin selection, onsite Testing and Integration.

Declaration:

I hereby declare that the Information above is true to the best of my Knowledge.

Dinesh R